L2.7 Newton's Method

§3.8 — use the formula, then decide where to start it

A calculator is the right tool for the arithmetic below, and the explorer in the lecture document is the right tool for watching *how* the method moves. Neither one draws these curves for you — the sketches are yours.

Newton's method

1. **A cubic, from a first guess you are given.** Use Newton's method with $x_1 = -1$ to find x_2 for the equation $x^3 + x + 3 = 0$.

   Then explain how the method works by sketching f and the tangent line at $(-1, 1)$, and marking x_2 on your sketch.

2. **A quartic.** Use Newton's method with $x_1 = 1$ to find x_2 for the equation $x^4 - x - 1 = 0$.

3. Nothing here is a polynomial. Use Newton's method to find the positive root of $\sin x = x^2$ correct to three decimal places.

Radians. Before you start: what is f, and what is f'?

> **Stop.** Nothing in the derivation of the formula ever mentioned the word *polynomial.* Which line of it, if any, would have to change here?

When Newton's method fails

4. **First show it is there, then go get it.** Consider the equation $3x^4 - 8x^3 + 2 = 0$ on the interval $[2, 3]$.

   (a) Explain how we know the equation must have a root in that interval. Name the theorem.

   (b) Approximate the root correct to four decimal places.

   Part (a) needs no calculator at all. Part (b) is where you choose x_1 — say why you chose the one you did.

At the boards

5. **BOARDS** **The Babylonian square-root rule.** Apply Newton's method to the equation $x^2 - a = 0$ to derive the algorithm

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right).$$

There is no number anywhere in this. Nothing is being approximated, and nothing is being evaluated — it is algebra, and the answer is a formula that computes $\sqrt{a}$ for *every* a at once. The Babylonians used it four thousand years before anyone drew a tangent line.

6. **BOARDS** **Division without dividing.** Apply Newton's method to the equation $\frac{1}{x} - a = 0$ to derive the algorithm

$$x_{n+1} = 2x_n - ax_n^2,$$

which computes $\frac{1}{a}$ using only multiplication and subtraction. This is how a machine that cannot divide finds a reciprocal.

This one is on tonight's homework as well — it is the same derivation with one term changed, so treat it as the rehearsal, not as new work.